Updated Project Scope and Objectives

07.Nov 2025

The revised focus of this project is centered on **hardware–software co-design** for distributed artificial intelligence (AI) execution across a fleet of drones and edge computing devices. The aim is to develop an **efficient architectural and algorithmic framework** that enables the partitioning and scheduling of deep learning models across heterogeneous hardware resources (drones) with limited onboard computational capabilities and nearby edge servers with higher processing capacity.

A key component of this research is the **design and evaluation of an intelligent model-splitting algorithm**. This algorithm should determine the optimal way to distribute neural network layers or computational blocks among multiple drones and edge nodes, based on factors such as hardware constraints (e.g., memory, processing power, and energy), communication bandwidth, and real-time scheduling deadlines. The objective is to ensure that offloading and distributed inference do not introduce unacceptable latency or communication overheads that violate system timing constraints.

The project will also involve **benchmarking and performance analysis** of the proposed framework. This includes quantifying the trade-offs between computation and communication costs when neural network operations are executed collaboratively among drones. For example, if Drone 1 performs an early convolutional layer and must transfer its intermediate feature maps to Drone 2 for subsequent layers, the system must evaluate whether the resulting communication delay (e.g., 5 ms) is acceptable within the target inference deadline or if it significantly degrades performance.

Furthermore, the research aims to prototype a **tool or simulation environment** that can automatically analyze and recommend feasible model-splitting strategies. Given a neural network and information about the available hardware resources, the tool should determine whether the model can be effectively partitioned and executed across the available drones and edge devices. It should also assess **schedulability**, indicating whether the distributed inference can meet timing requirements under specific operational conditions, such as when drones are capturing and processing field data in real time.

Ultimately, while **model-splitting techniques** have been explored in prior research, this project seeks to **advance the field** by developing a more intelligent, adaptive, and architecture-aware method that incorporates both algorithmic feasibility and system-level performance constraints. The study will also explore the **communication protocols and architectural design**

considerations required to enable such coordinated AI inference in drone swarms.

The overarching research questions include assessing the **feasibility and efficiency** of distributed model execution in drone–edge ecosystems and determining whether the proposed approach can lead to measurable improvements in latency, throughput, and energy efficiency compared to existing methods.